Business Question and Visualization Report

Date	24 September 2024
Team ID	
Project Name	EV Market Analytics Dashboard
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. Which vehicle categories contribute the most to annual sales?

Visualization: Bar chart comparing total annual sales by vehicle category (hatchback, sedan, SUV, bus, truck, two-wheeler).

[Screenshot of visualisation]

2. How have total EV sales changed year over year?

Visualization: Line chart showing total annual sales by year with year-over-year growth % shown in the tooltip.

[Screenshot of visualisation]

3. Which manufacturers hold the largest market share?

Visualization: Treemap (or donut chart) showing market share % of the top manufacturers, with a bar chart of total sales.

[Screenshot of visualisation]

4. How does customer satisfaction vary across vehicle categories?

Visualization: Bar chart showing average customer rating by vehicle category, coloured by rating level.

[Screenshot of visualisation]

5. Which battery technologies are most widely adopted?

Visualization: Pie / donut chart showing the share of sales by battery type (LFP, NMC, NCA, Sodium-Ion, Solid-State).

[Screenshot of visualisation]

6. How does driving range compare across battery types and vehicle categories?

Visualization: Heat map (or grouped bar chart) showing average driving range by battery type and vehicle category.

[Screenshot of visualisation]

7. Which countries lead EV manufacturing?

Visualization: Filled map showing manufacturing countries, shaded by total sales, with a supporting bar chart.

[Screenshot of visualisation]

8. Is a larger battery capacity linked to a longer driving range?

Visualization: Scatter plot of battery capacity vs driving range, coloured by battery type, with a trend line.

[Screenshot of visualisation]